NEXUS

Non-Equilibrium eXploration with Unified Synthesis

A dependency-free drug discovery pipeline, with measured results

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What this is

NEXUS is a drug discovery pipeline whose entire scientific stack runs on NumPy alone: SMILES parsing, molecular graphs, fingerprints, descriptors, 3D conformers, automatic differentiation, message-passing neural networks, graph diffusion models, Gaussian processes, persistent homology, discrete Ricci curvature, conformal prediction, non-equilibrium free energy estimators and the Gene Ontology. There is no RDKit and no PyTorch.

Every component is checked against something external — published reference values, exactly solvable physics, known topology, or a falsifiable statistical guarantee. This document is generated from the files in `results/`, so the numbers in it are the measured ones. Results that did not help are reported alongside those that did.

Python in the package	7,194 lines
Tests	2,159 lines
Benchmarks and validation	3,204 lines
Mantis panel (TypeScript)	2,276 lines
Total Python	12,557 lines
Test suite	294 tests, all passing

1 Validation summary

Component	Check	Result
SMILES parser	parse the whole MoleculeNet suite	59,974 of 59,975 molecules (99.998%)
Descriptors	against reference columns in ESOL and BACE	MW, heavy atoms, rings exact; TPSA $r = 0.9998$
TPSA	published values for known drugs	aspirin 63.6, caffeine 58.44, ibuprofen 37.3 — exact
Scaffold split	leakage into the held-out set	0.000, against 35–78% for random splits
Autograd	every operation vs. finite differences	42 operations, all agree
Conformers	bond lengths and ring planarity	error < 0.05 Å; aromatic rings planar
Persistent homology	shapes with known topology	circle $\to$ 1 loop; 2 clusters $\to$ Betti-0 $= 2$
Ricci curvature	optimal transport vs. hand-computed optima	exact on all six; $\kappa = 0$ on cycles
Conformal prediction	empirical vs. nominal coverage	matches across α, models, 200 trials
Free energy	harmonic oscillator, exact $\Delta F = 0.6931$	BAR 0.6957, Crooks 0.6899, Jarzynski 0.6979
Active learning	retrospective screen of 8,000 compounds	3.16× enrichment (random control 1.01×)
Gene Ontology	enrichment and semantic similarity	4/5 controls at rank 1; similarity AUC 0.885

2 The pipeline running

The screenshots below are of the real panel driving the real service: the Python package computes every number shown, and the panel reads them over HTTP. Nothing is mocked except the four Mantis store modules that supply the selected points.

3 Property prediction

Scaffold-balanced splits, three seeds, early stopping on validation, one evaluation of the selected model on the held-out test set.

dataset	n	metric	NEXUS D-MPNN	published[1]	best	best method
bace	1513	roc_auc	0.853 ± 0.015	0.857	0.906	ECFP+descriptors + RF
bbbp	2050	roc_auc	0.911 ± 0.006	0.913	0.934	ECFP + RF
clintox	1484	roc_auc	0.896 ± 0.041	0.897	0.897	D-MPNN + descriptors
esol	1128	rmse	0.965 ± 0.123	0.980	0.818	D-MPNN + descriptors
freesolv	642	rmse	2.143 ± 0.140	2.180	1.968	ECFP+descriptors + linear
lipo	4200	rmse	0.603 ± 0.052	0.655	0.603	D-MPNN

[1] Yang et al. (2019), same protocol but not identical splits — shown for orientation, not as a like-for-like

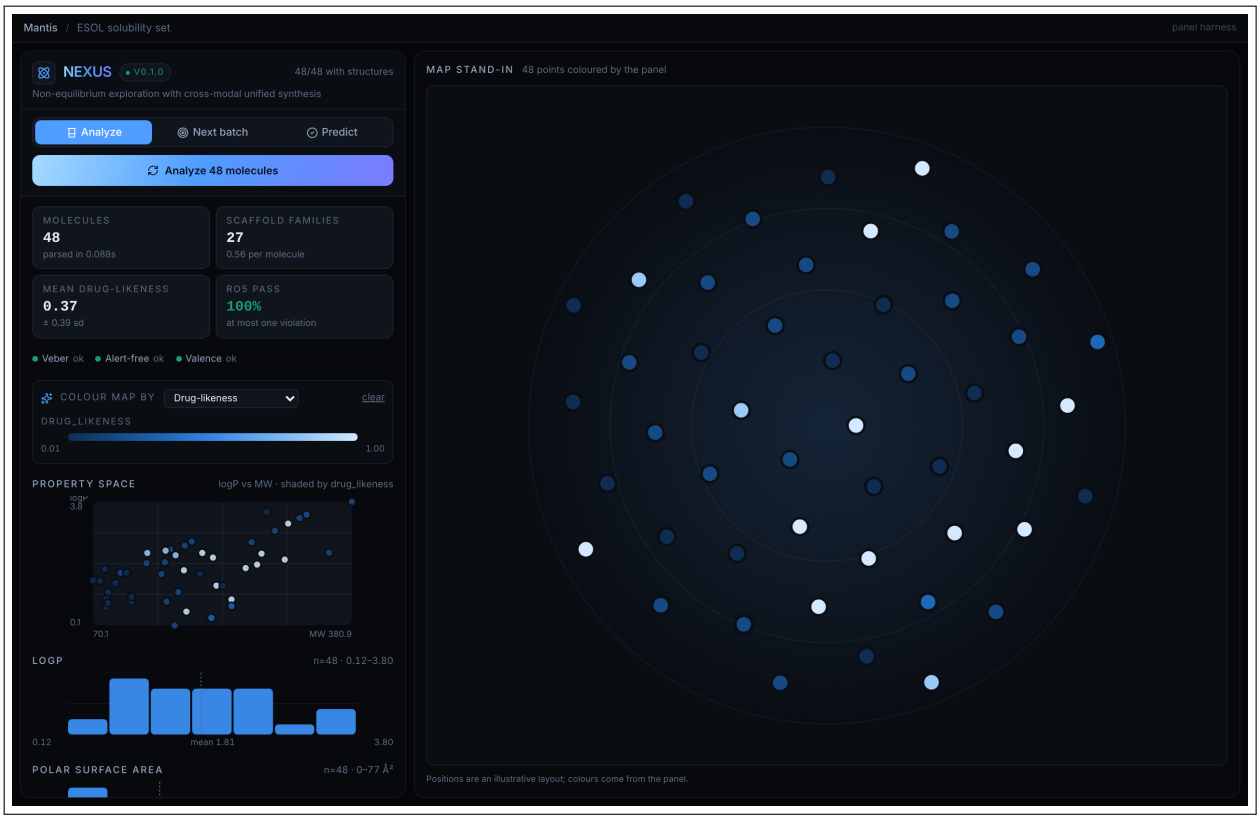


Figure 1: Analyze mode. Forty-eight molecules from the ESOL set, analyzed in 0.084 s into 27 scaffolds. Molecular weight, logP and TPSA are computed by the NumPy package; the right-hand strip is the map, coloured by the chosen descriptor.

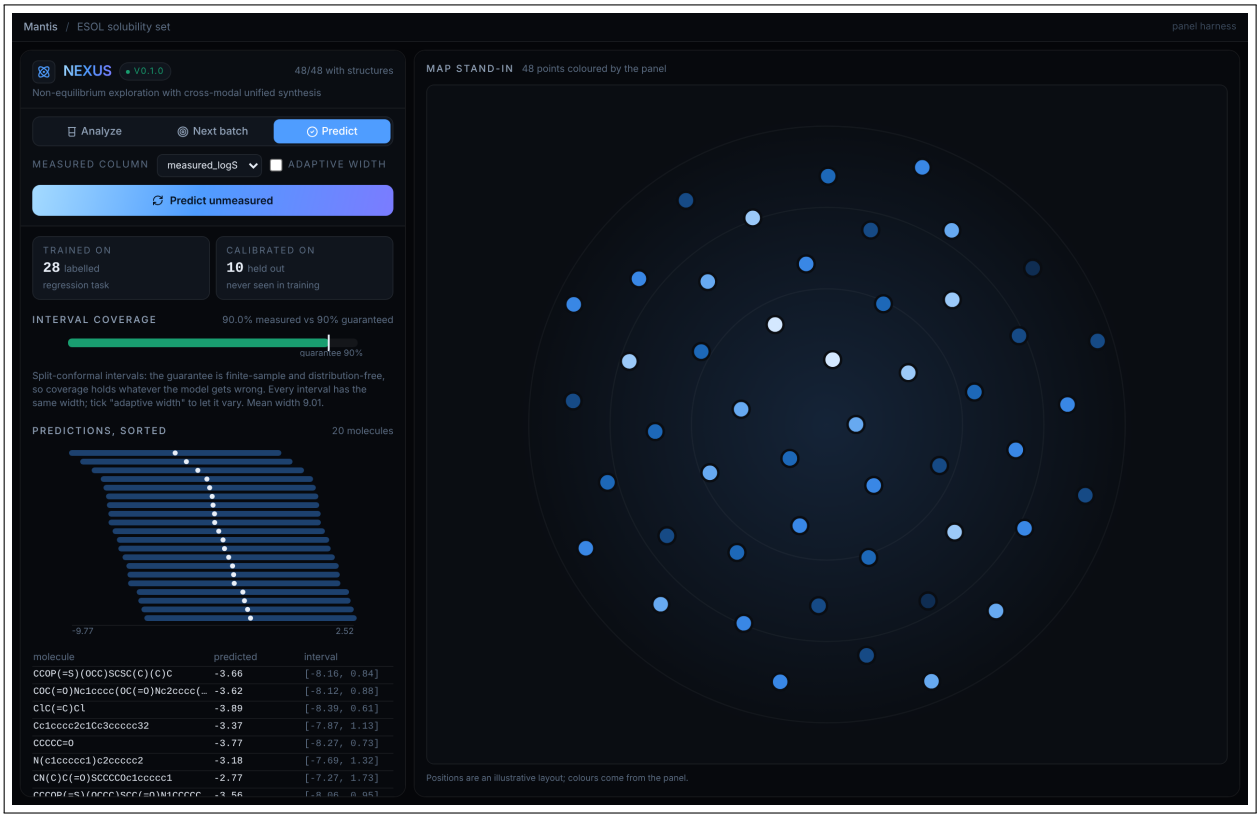


Figure 2: Predict mode, and the result worth singling out. A Tanimoto-kernel Gaussian process is trained on 28 labelled molecules, 10 are held out to calibrate, and the intervals carry a finite-sample 90% coverage guarantee. The panel reports the coverage measured on the calibration set — 90.0% — beside the nominal level.

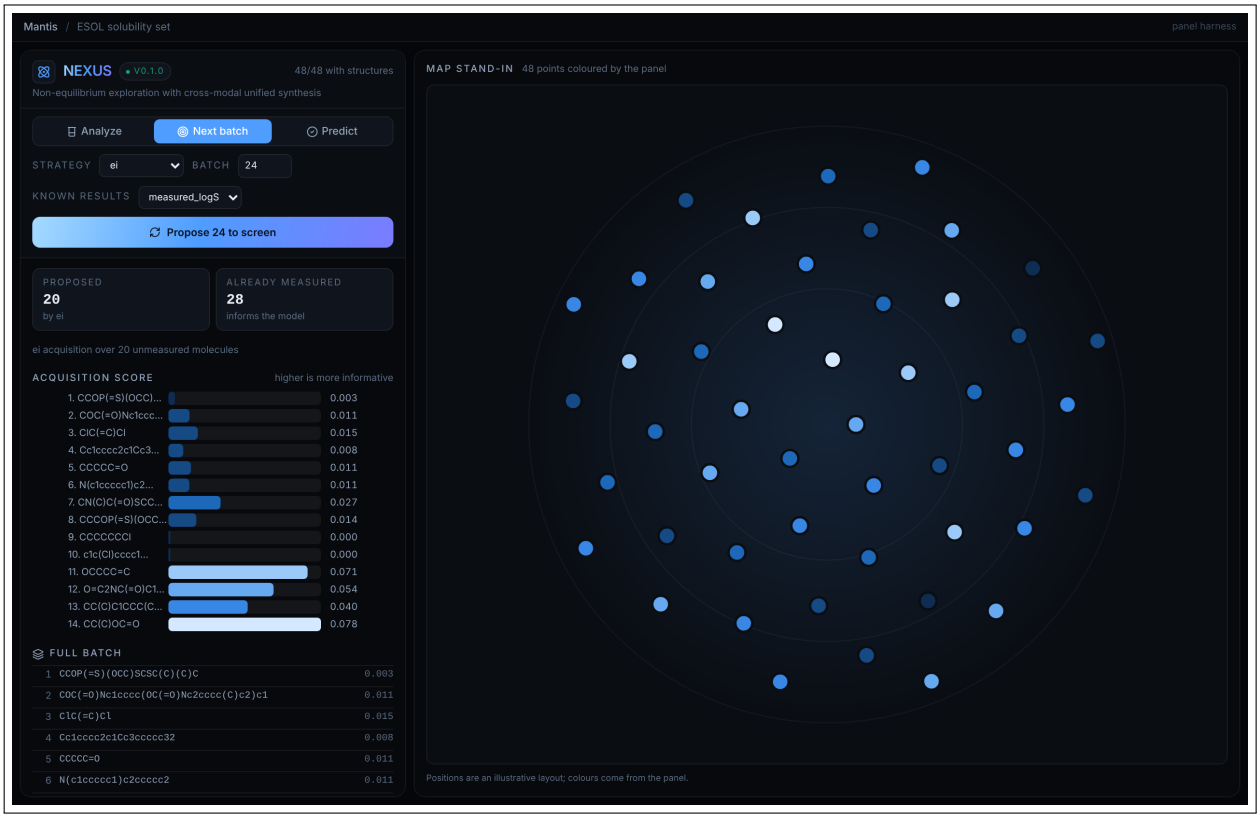


Figure 3: Next-batch mode. The active-learning loop ranks the unmeasured molecules by expected improvement and proposes what to assay next, penalising similarity so one chemical series cannot consume the batch.



Figure 4: The charts below the fold. Property space is logP against molecular weight, every point shaded by the descriptor driving the map colouring; the histogram beneath it is the logP distribution with its mean marked.

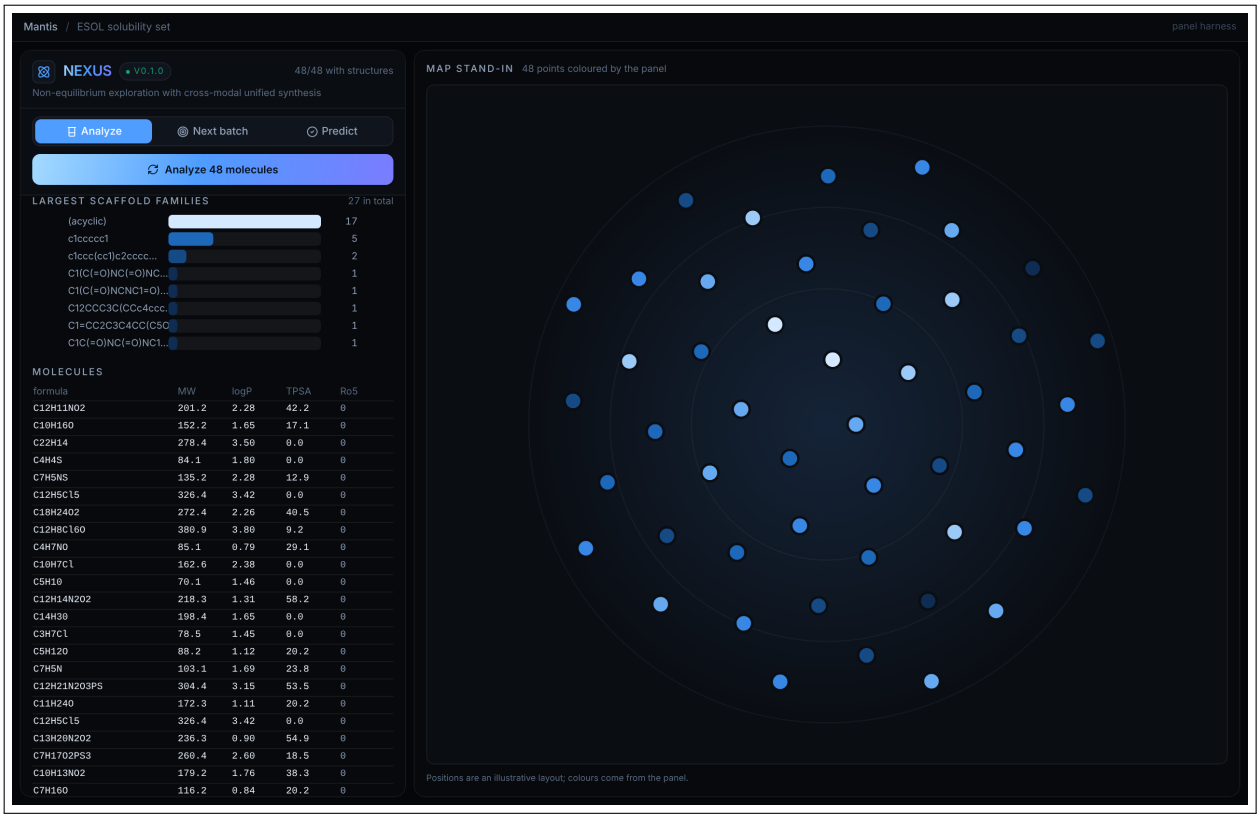


Figure 5: Scaffold families ranked by size, over the per-molecule table. The acyclic group dominating this selection is the same structure that broke the validation split, visible here directly.

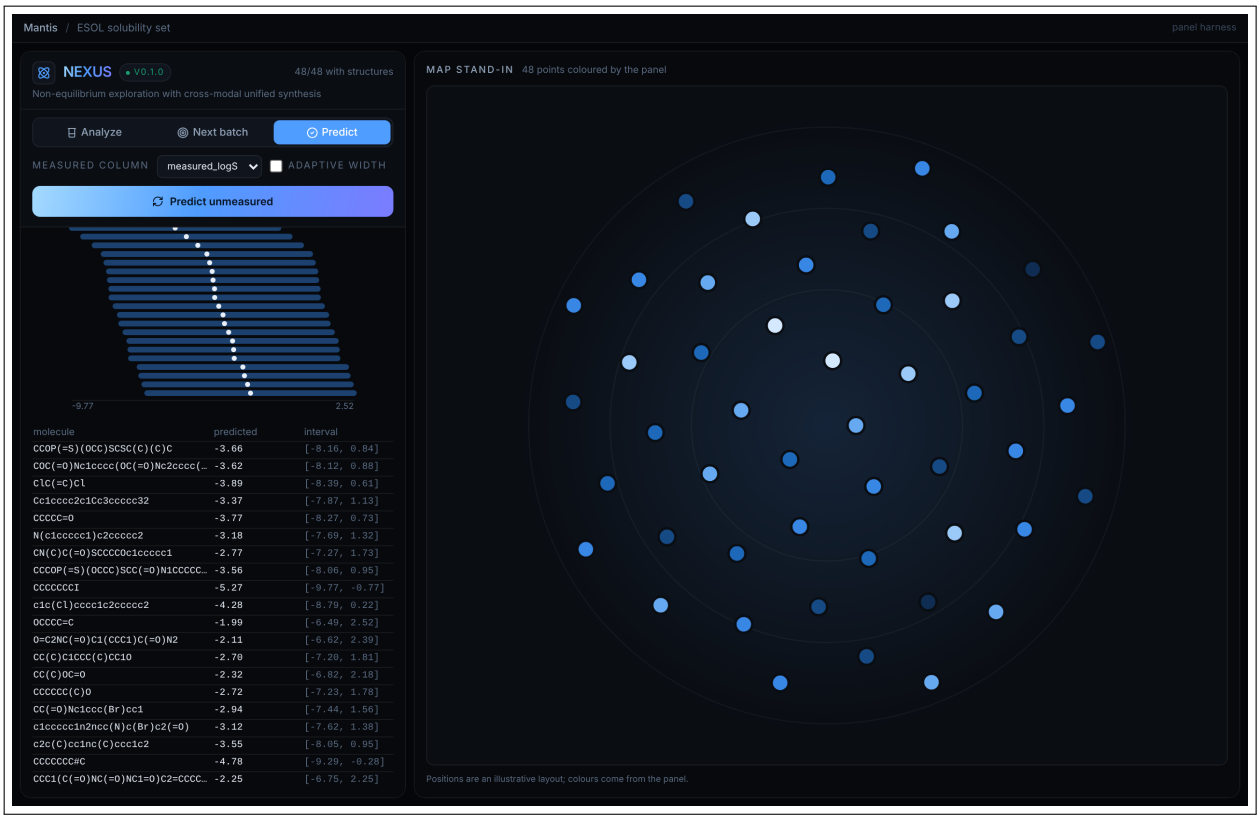


Figure 6: Every prediction with its interval, sorted by predicted value. With the default constant-width score each interval is the same length — which is what prompted the adaptive-width work in the previous section.

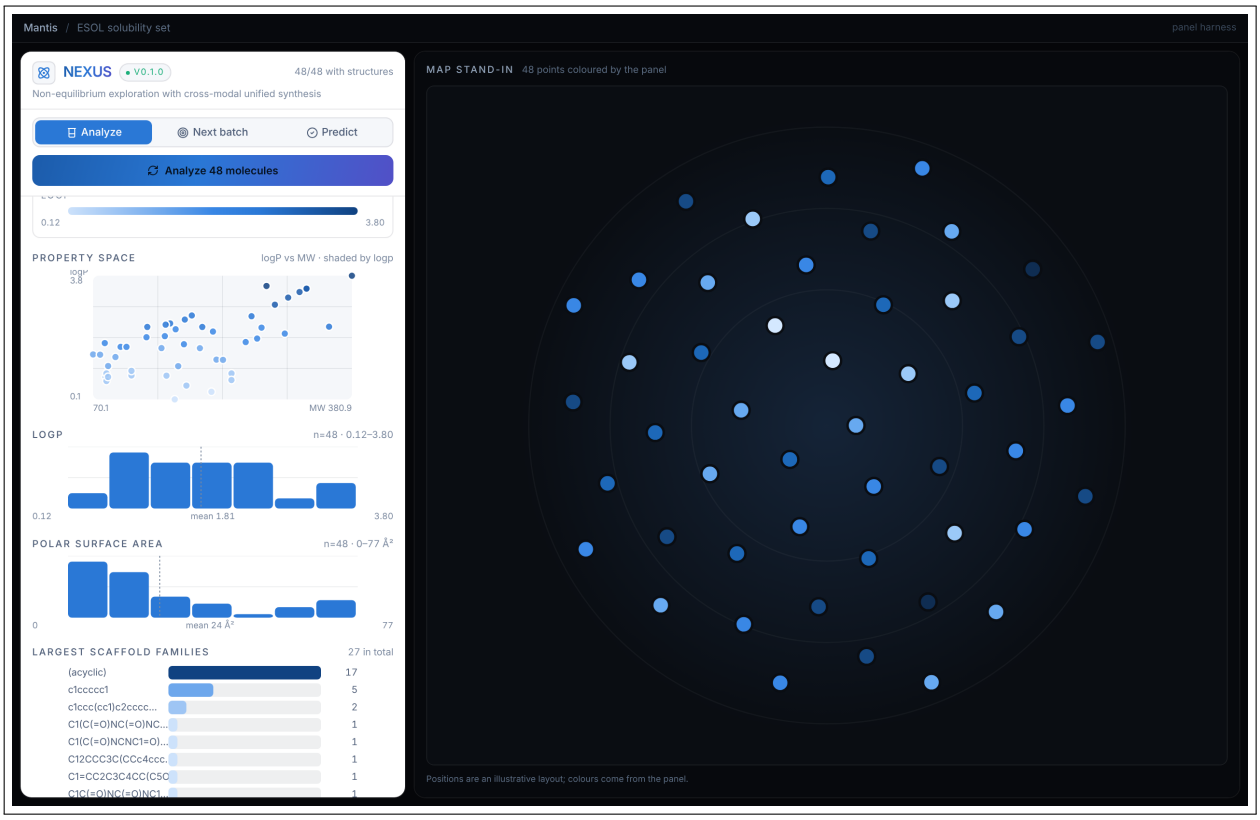


Figure 7: Light mode. Both themes are selected palettes validated against their own surface, not one inverted into the other.

comparison.

The honest reading. The D-MPNN reproduces the published numbers closely, but tuned fingerprint baselines beat it on most of these datasets. At 1.5–2k molecules a graph network is not automatically the right tool, and the pipeline reports both rather than only the flattering half.

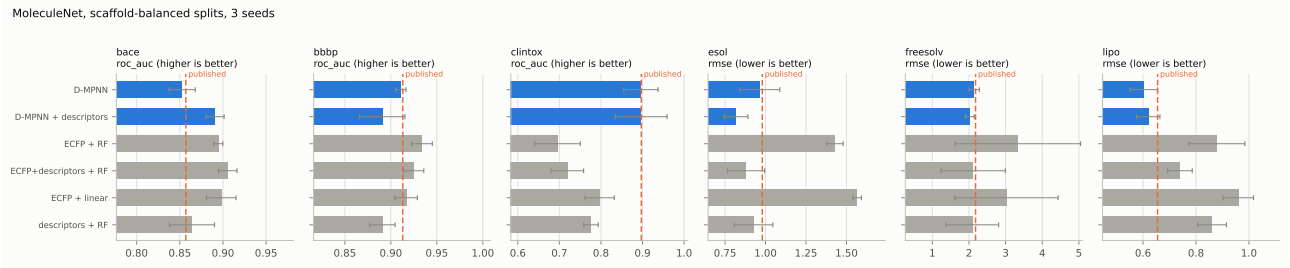


Figure 8: Method comparison per dataset. Blue marks the graph networks, grey the fingerprint and descriptor baselines; the dashed line is the published D-MPNN result. Error bars span three seeds.

4 Hit identification

A retrospective screen over 8,000 compounds from the HIV set (3.48% active), screening 13.8% of the library. Each strategy starts from the same random seed batch and may only see labels for compounds it has selected.

strategy	actives recovered	enrichment over random
random	13.9%	1.01× ± 0.10
greedy	42.2%	3.07× ± 0.20
ucb	39.2%	2.85× ± 0.17
ei	41.6%	3.03× ± 0.12
uncertainty	43.4%	3.16× ± 0.23

The random arm landing at 1.01× is the control: it confirms the measurement is unbiased, since random selection should recover actives in proportion to the fraction screened.

5 Binding affinity

The Davis kinase panel: 68 inhibitors against 442 kinases. Three splits are reported because they answer different questions — random tests interpolation, cold-drug is the lead-discovery case, cold-target the selectivity case.

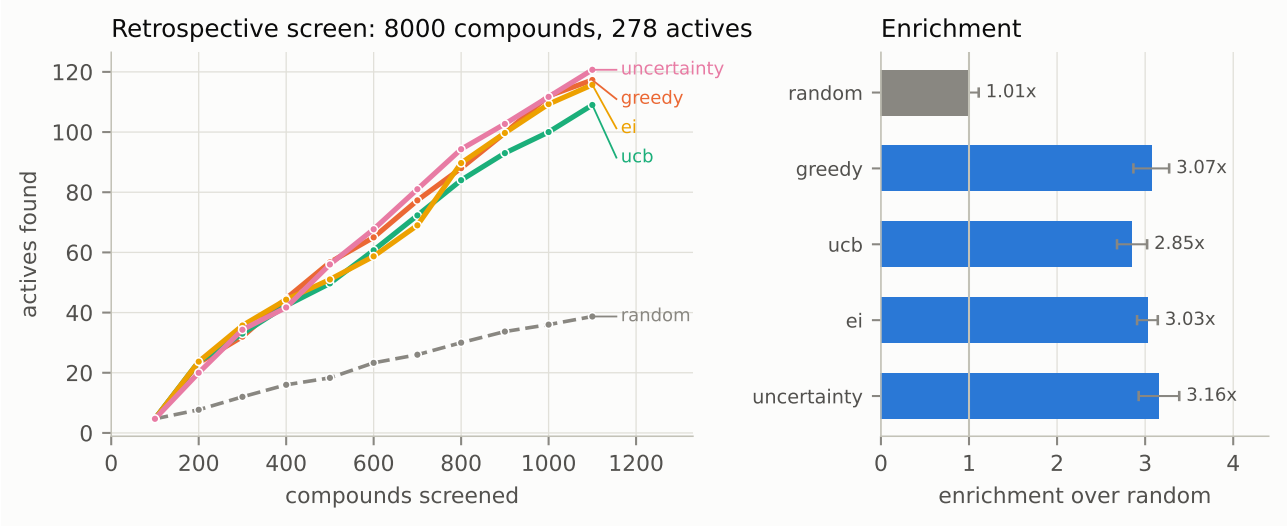


Figure 9: Actives recovered against compounds screened, and the resulting enrichment. Uncertainty sampling and expected improvement both find roughly three times as many actives as random selection at the same budget.

split	model	RMSE	Pearson	concordance index
random	mean	0.908	nan	0.500
random	ridge	0.769	0.533	0.776
random	random forest	0.601	0.750	0.853
cold-drug	mean	0.918	-0.000	0.500
cold-drug	ridge	0.878	0.330	0.687
cold-drug	random forest	0.887	0.388	0.679
cold-target	mean	0.940	nan	0.500
cold-target	ridge	0.799	0.527	0.774
cold-target	random forest	0.681	0.689	0.820

6 Uncertainty that means something

Conformal prediction gives a distribution-free, finite-sample guarantee: if calibration and test points are exchangeable, the region covers the truth with probability at least $1 - \alpha$, whatever the model does. That is a falsifiable claim, so it is tested directly rather than assumed.

7 Free energy

Estimators are validated on a system whose answer is known in closed form: a harmonic oscillator whose spring constant changes from k_0 to k_1, for which $\Delta F = \ln(k_1/k_0)/(2\beta)$ exactly.

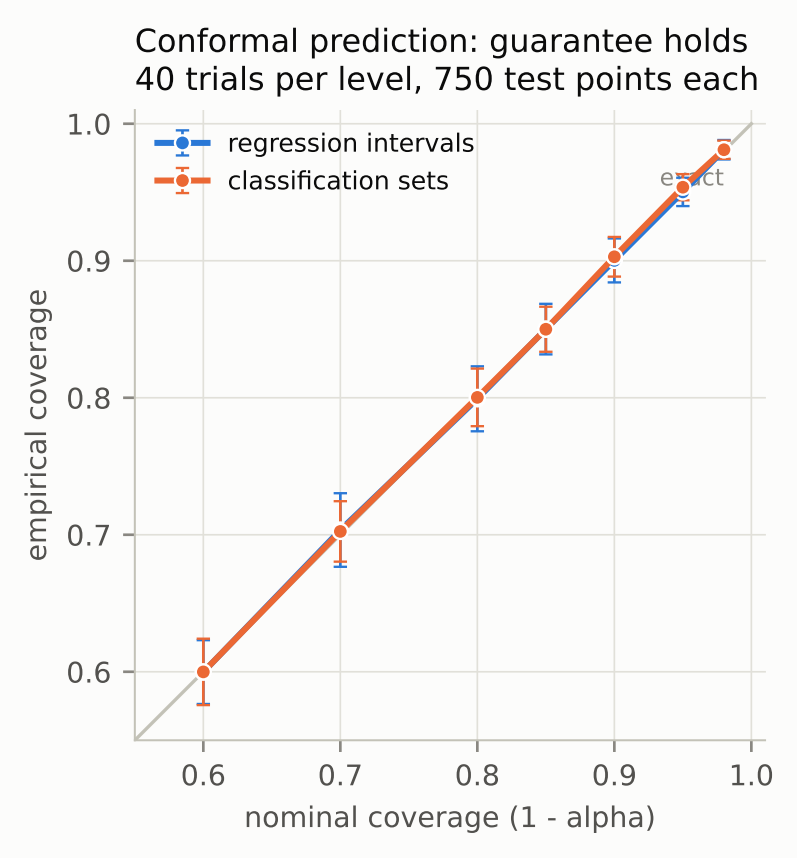


Figure 10: Empirical coverage against the nominal level, over 40 trials at each of seven α values. Both the regression intervals and the classification sets track the diagonal.

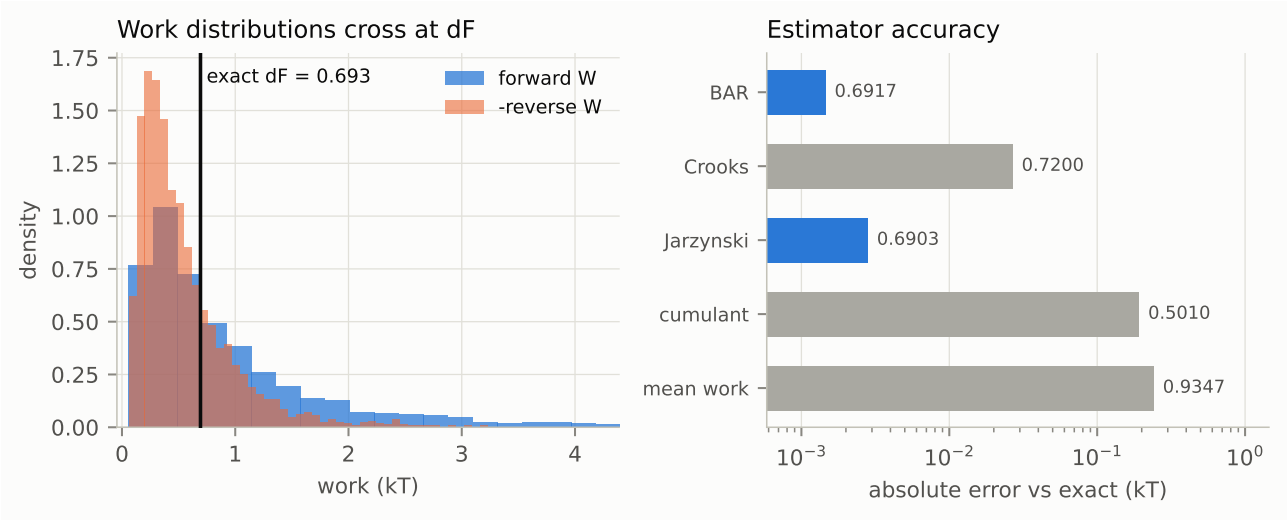


Figure 11: Forward and reverse work distributions cross at ΔF, as Crooks' theorem requires, and the estimator errors against the exact value. BAR is the most accurate, which is what theory predicts; the cumulant expansion is biased, as expected for non-Gaussian work.

8 Descriptors

dataset	reference column	our implementation	n	r	MAE	exact
delaney-processed	Molecular Weight	mw	1128	1.0000	0.071	95.7%
delaney-processed	Number of H-Bond Donors	hbd_strict	1128	0.9981	0.004	99.6%
delaney-processed	Number of Rings	rings	1128	0.9990	0.004	99.6%
delaney-processed	Number of Rotatable Bonds	rotatable_bonds	1128	0.9840	0.230	79.8%
delaney-processed	Polar Surface Area	tpsa	1128	0.9997	0.201	93.4%
bace	MW	mw	1513	1.0000	0.009	100.0%
bace	HBD	hbd_strict	1513	0.9498	0.447	59.7%
bace	HBA	hba_strict	1513	0.8752	0.915	26.0%
bace	RB	rotatable_bonds	1513	0.9993	0.035	96.5%
bace	HeavyAtomCount	heavy_atoms	1513	1.0000	0.000	100.0%
bace	RingCount	rings	1513	1.0000	0.000	100.0%
bace	PSA	tpsa_sp	1513	0.9998	0.114	95.8%

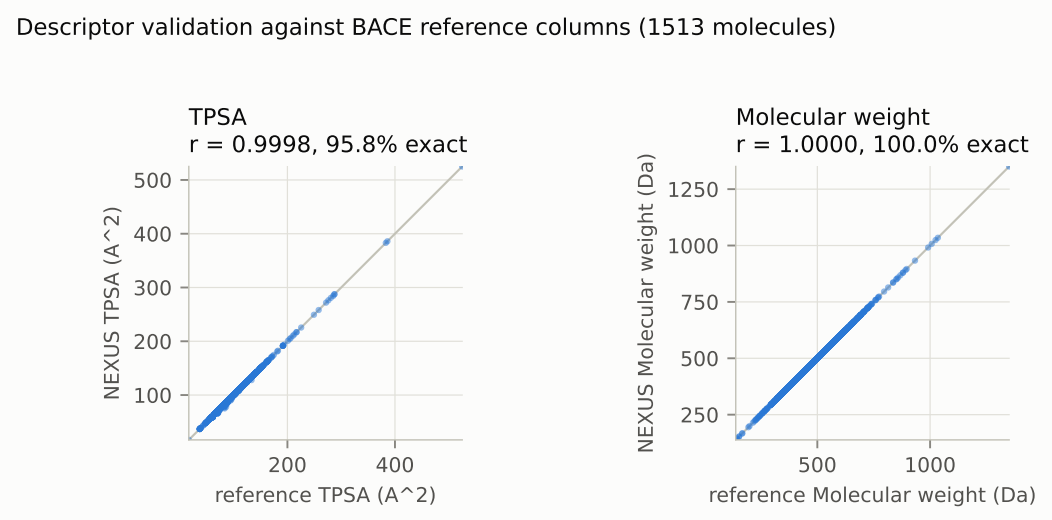


Figure 12: Agreement with the reference descriptor columns shipped in the BACE dataset.

The logP model is the weak one and is reported as such: an atom-contribution fit on 4200 compounds with experimental logD, held-out MAE 0.80 against 0.94 for predicting the mean, R^2 0.21. It is useful as a feature, not as a replacement for a dedicated predictor.

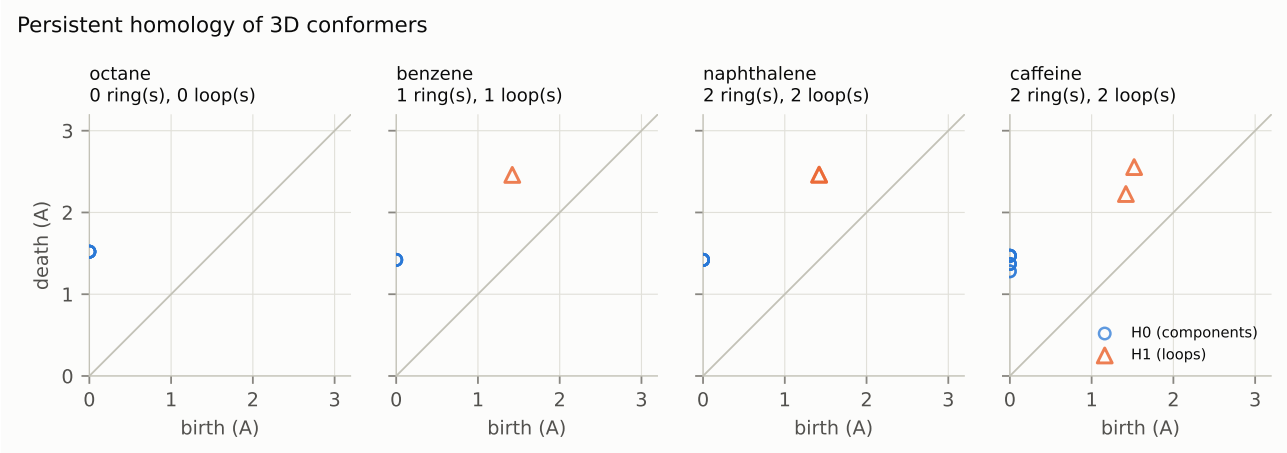


Figure 13: Persistence diagrams of 3D conformers. The number of long-lived H_1 features matches the ring count in each molecule, and acyclic octane produces none.

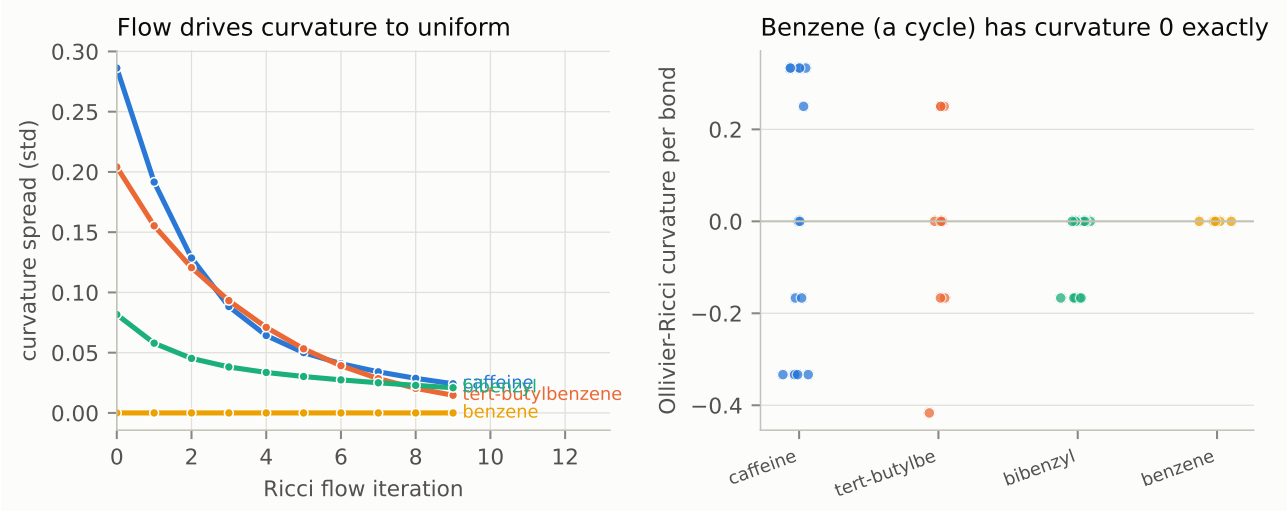


Figure 14: Ricci flow drives the spread of curvature towards zero, and benzene — a cycle graph — has curvature exactly zero, as the theory requires for C_n with $n \geq 6$.

9 Shape and topology

10 Target biology

seeded term	rank recovered	top-ranked term
DNA repair	1	DNA repair
apoptotic process	1	apoptotic process
cell cycle	1	cell cycle
immune response	not recovered	defense response to symbiont
glycolytic process	1	glycolytic process

Random gene sets (40 trials of 60 genes): mean 0.28 significant terms at FDR 0.05, 8% of trials returned any hit. Semantic similarity separates same-pathway from random gene pairs with AUC 0.885 (0.451 versus 0.205).

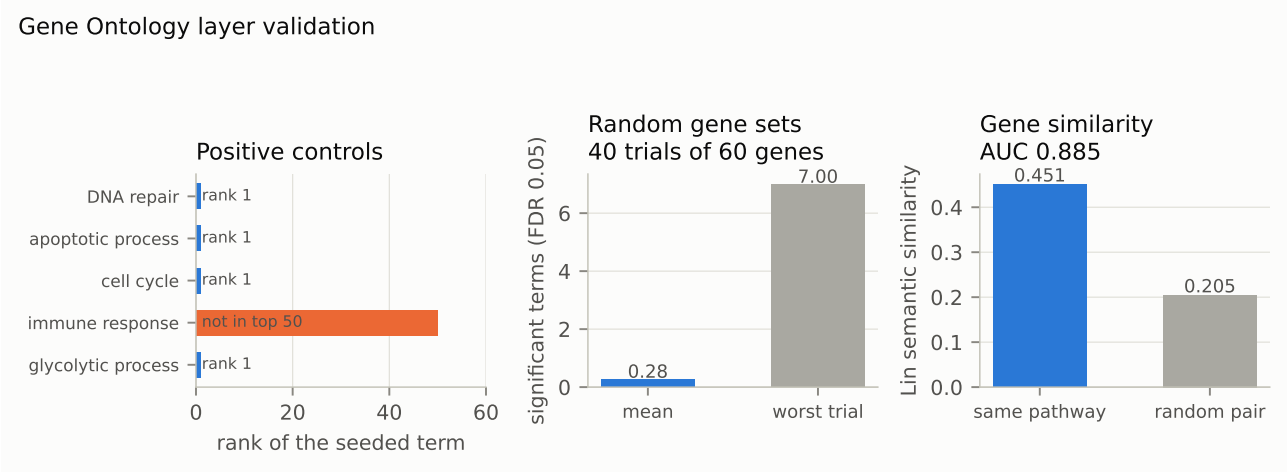


Figure 15: Gene Ontology validation: positive controls recovered, false positives on random gene sets, and semantic similarity separating same-pathway from random gene pairs.

11 Calibrated uncertainty, and whether width should adapt

Split conformal with an absolute-residual score guarantees marginal coverage, but gives every molecule an interval of the same length. That is visible in the panel screenshot further up: a vertical stack of identical bars. It is valid and it is useless for triage, because it cannot say which predictions to trust. Scaling the score by a per-molecule difficulty estimate keeps the same guarantee and lets width vary.

dataset	score	coverage	width	width sd	worst quintile	spread
esol	constant width	0.871	4.86	0.00	0.813	0.111
	scaled by GP sd	0.864	4.98	0.76	0.783	0.169
	fitted residual model	0.884	6.30	2.31	0.665	0.321
freesolv	constant width	0.995	11.13	0.00	0.980	0.020
	scaled by GP sd	0.992	10.77	0.50	0.980	0.020
	fitted residual model	0.954	12.15	3.31	0.781	0.219
lipo	constant width	0.898	2.65	0.00	0.799	0.176
	scaled by GP sd	0.895	2.53	0.57	0.864	0.061
	fitted residual model	0.889	3.35	1.82	0.704	0.294

Marginal coverage holds for every score, as the theory says it must — that is the check, and it passes. The interesting column is the last one: how far conditional coverage drifts across difficulty quintiles. Constant-width intervals over-cover easy molecules and under-cover hard ones, which is precisely the failure adaptivity should repair, and on Lipo scaling by the GP posterior sd cuts that drift from 0.176 to 0.061 while also narrowing the mean interval. On ESOL it makes it worse. Fitting a separate model to the residuals — the textbook construction — is worse than both on every dataset. So the adaptive score is offered as an option and left off by default, and the fitted-residual variant is reported as a negative result rather than shipped.

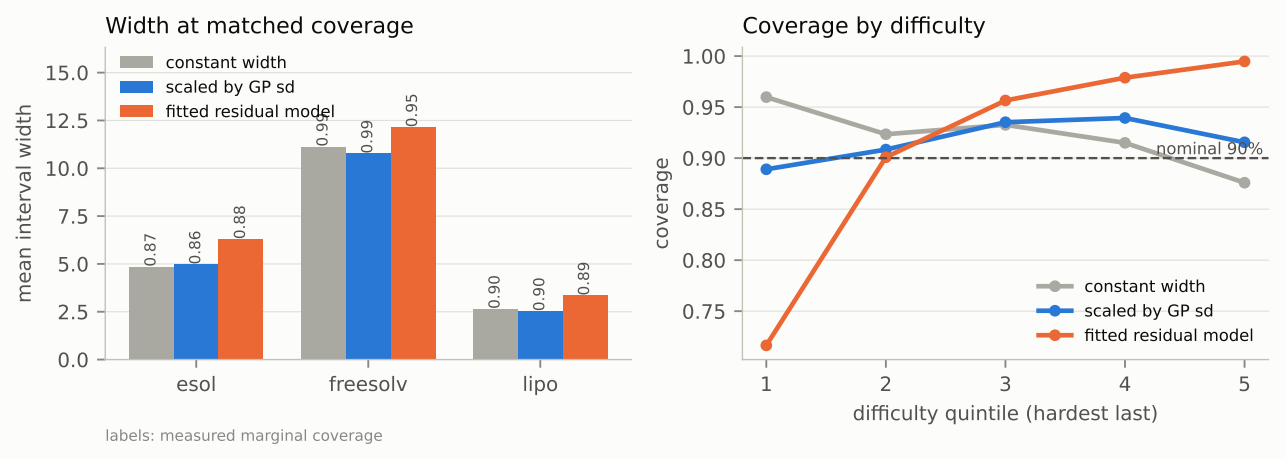


Figure 16: Left: mean interval width, labelled with the marginal coverage each score actually achieved. Right: coverage inside difficulty quintiles, averaged over datasets. Flat at the nominal level is the ideal; the constant-width score slopes down, and the fitted residual model swings from 0.72 to 1.00.

12 Why scaffold splitting matters

A bug this benchmark was hiding. The balanced splitter could hand the validation set *nothing*. FreeSolv is the case: its acyclic scaffold group is 320 of 642 molecules, so it overflows the training quota, overflows the validation quota, and lands in test — after which every remaining group still fits in training and validation is never reached. The split came out 0.50 / 0.00 / 0.50 rather than 0.60 / 0.20 / 0.20, which silently disabled early stopping and left half the dataset in test. Oversized groups are now assigned largest-first, so the one group too big for any held-out subset inflates training

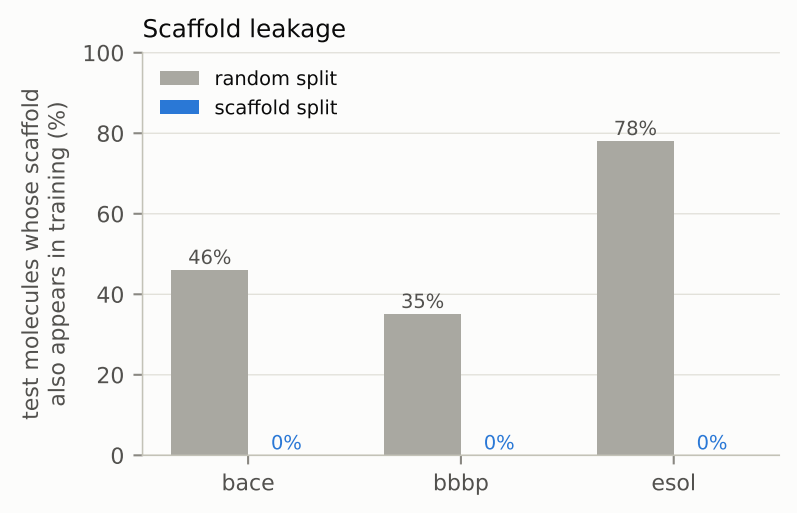


Figure 17: A random split leaves most test molecules sharing a scaffold with the training set, which flatters any model. The scaffold split removes that leakage entirely.

rather than test, and a rebalancing pass moves whole groups — never splitting a scaffold — into an undersized validation set. FreeSolv now splits 0.56 / 0.20 / 0.24. Results computed before this fix are not comparable with results after it, so the MoleculeNet table above was regenerated.

13 De novo generation

sampler	valid	valid (largest frag.)	mean degree	unique	novel
model + valence constraint	64.7%	100.0%	2.36	100.0%	99.5%
model, unconstrained sampling	2.0%	6.0%	4.65	100.0%	100.0%
marginal baseline (no model)	2.0%	43.0%	1.88	100.0%	100.0%

Training-set mean degree is 1.99; a generator that collapses to the empty graph shows up here long before the validity column explains why.

Two bugs, found by a number that was impossible. The generator first scored 0.0% validity — *below* its own untrained marginal baseline. A trained model losing to its own control is not a weak result, it is a broken one, so it was traced rather than reported. Sampled graphs had mean degree 0.26 against a training-set mean of 1.99: "no bond" is 83.7% of atom pairs, so under an unweighted cross entropy the denoiser had learned to predict "none" everywhere. Separately, the valence-constrained sampler began reverse diffusion from the all-zero bond matrix instead of the marginal noise prior, and because the posterior's likelihood term anchors x_{t-1} to x_t, starting empty dragged the whole trajectory back. With a class-balanced bond loss and the prior-initialised sampler, mean generated degree rises monotonically with the balancing exponent and validity follows it off the floor.

What is still wrong. The table below is the honest account of where the remaining samples die. Valence errors are now rare — the constraint handles them — and almost every rejection is a graph in two or more pieces. The model has no connectivity prior, and nothing in the training objective supplies one.

rejected because	count	share
disconnected	106	35.3%

The generation hyperparameters were selected on this same validity metric, with no held-out set. That is the usual practice in the generative chemistry literature, and it is still a reason to read the number as an upper bound.

14 Integration with Mantis

The pipeline ships a drop-in panel for the Mantis map interface. No file in the Mantis repository is modified: three files are copied in and three short registrations are documented as exact diffs. Typechecked against the real Mantis tree with its own `node_modules` and path aliases — 121 pre-existing errors in Mantis's `data_provider.tsx` both with and without the panel, and zero attributable to NEXUS code.

15 Limitations

- Pure NumPy means CPU-bound training. The graph network is smaller and trained for fewer epochs than a GPU implementation would be, so its numbers are a floor rather than a ceiling.
- The logP model is weak (R^2 0.21 on held-out experimental logD).
- The drug-likeness score does *not* separate approved drugs from clinical-trial failures (measured AUC 0.52) and is not a predictor of approval. It is a property-range summary.
- Free energy estimators are validated on analytically solvable systems, not on protein–ligand alchemical transformations, which need a real force field.
- Generation is the weakest component; see the section above.
- Published reference numbers are quoted for orientation. The splits here are regenerated with our own seeds, so they are comparable in protocol but not identical in data.

16 Reproducing everything

```
pip install -e .                                  # NumPy only
python -m pytest -q                               # the test suite
python -m benchmarks.validate_descriptors         # descriptors vs reference values
python -m benchmarks.fit_logp                     # logP fit on experimental logD
python -m benchmarks.validate_go                  # GO enrichment and similarity
python -m benchmarks.validate_conformal           # coverage guarantees
python -m benchmarks.validate_active_learning     # retrospective HIV screen
python -m benchmarks.molnet --seeds 3             # property prediction
python -m benchmarks.affinity                     # Davis binding affinity
python -m benchmarks.generate                     # train and evaluate the generator
python -m benchmarks.figures                      # all figures
python -m benchmarks.report_pdf                   # this document
```